

Ayush Sharma

Associate Software Engineer | Bengaluru, India

+91 7006911028 ayush.nitsgr@gmail.com sharma-ayushhh ayuxharma ayush-sharma.io LeetCode

Professional Summary

Software Engineer building reliable backend, distributed, and cloud-based systems in C++ and Python. Experienced with production Oracle Integration Cloud pipelines, REST APIs, automation, and root-cause analysis across the SDLC. Built a Raft-based replicated key-value store with gRPC and an explainable API-failure diagnostic combining deterministic rules with LangGraph-based LLM fallback.

Technical Skills

Languages: Python, C++, SQL

Testing & QA: Test Planning & Execution, API Testing, Debugging, Defect & Root-Cause Analysis, Log Analysis, CTest

Gen AI: Generative AI, LLM Integration (Gemini API), LangGraph, Agentic Workflows, Prompt Design

APIs & Protocols: REST APIs, FastAPI, gRPC, Protocol Buffers, OAuth, JWT

Core Fundamentals: SDLC, Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Cloud & Tools: Oracle Integration Cloud (OIC), OCI Serverless Functions, Docker, Git

Work Experience

Oracle

Sept 2025 – Present

Associate Engineer, CSS Global SaaS & Apps Delivery

Bengaluru, India

- Designed, coded, tested, and deployed enhancements for **10+ integration suites** on **Oracle ERP Cloud**, collaborating with cross-functional project teams to deliver reliable data ingestion pipelines across distributed platforms.
- Built and tested production-grade Python **REST APIs** and automation utilities, executing structured test plans and debugging protocols to validate reliability and reducing manual operational overhead by **80%**.
- Identified, logged, and debugged recurring API rate-limit defects in production subsystems, implementing a fix through **Oracle Integration Cloud (OIC)** configuration that eliminated **15 daily failures**.
- Led **end-to-end disaster recovery (DR) activity**, ensuring operational resilience and reliability.
- Developed serverless **OCI functions** to support scalable architecture, improving downstream data reliability and cutting manual data-organization effort by **75%**.

Projects

Autonomous API Failure Debugger | Python, FastAPI, LangGraph, Gemini API, Docker [Live Demo](#) | [Source Code](#)

- Engineered a 7-stage **LangGraph** workflow that automates root-cause triage & defect analysis for distributed API failures.
- Built a rule-based classifier covering 9 failure categories using weighted keyword and stack-trace signal matching.
- Implemented hypothesis generation and weighted evidence-elimination logic with mutually exclusive hypothesis groups to narrow down root causes.
- Produced ranked, confidence-scored root causes and fixes using a hybrid **LLM + rule-based fallback architecture**.

MiniRaftKV – Distributed Key-Value Store | C++, Raft, gRPC, Protocol Buffers, CMake, CTest [Source Code](#)

- Built a 3-node distributed key-value store in **C++** using the **Raft consensus algorithm**, implementing leader election, randomized election timeouts, heartbeats, log replication, majority-based commits, and follower recovery.
- Developed a replicated **SET/GET/DELETE** state machine applying committed log entries exactly once for consistent data across nodes.
- Implemented **gRPC** and **Protocol Buffer** APIs for Raft communication (**RequestVote**, **AppendEntries**) and client operations, with thread-safe request handling and deadline-bound peer communication.

Education

National Institute of Technology, Srinagar

2021 – 2025

Bachelor of Technology in Electrical Engineering

CGPA: **8.48/10**

DAV School, Jammu

2019 – 2021

Central Board of Secondary Education

10th: **93.6%** | 12th: **92.4%**

Achievements

- Solved **400+** Data Structures & Algorithms problems on LeetCode.
- Campus Recruitment Coordinator – collaborated with **80+ companies** to streamline the hiring process.
- Selected for Yuva Sangam Phase IV Exchange Program with VNIT Nagpur (EBSB, Ministry of Education).